



THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

OBJECT ORIENTED PROGRAMMING

LAB Assignment 2

SUBMITTED BY:

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Batch: 2023

Question-1 →

```
#include <iostream>

#include <math.h>

#include <string> using
namespace std;


struct students{ string
name; long roll; string
degree; char hostel; float
cgpa; void addDetails(){
cout<<"Enter Name: ";
cin>>name; cout<<"Enter
Roll.No: "; cin>>roll;
cout<<"Enter Degree: ";
cin>>degree;
cout<<"Enter hostel: ";
cin>>hostel;
cout<<"Enter cgpa: ";
cin>>cgpa; } void
updateDetails(){
cout<<"Enter new Name:
"; cin>>name;
cout<<"Enter new
```

```
Roll.No: "; cin>>roll;

cout<<"Enter new

Degree: "; cin>>degree;

cout<<"Enter new hostel:

"; cin>>hostel;

cout<<"Enter new cgpa:

"; cin>>cgpa;

}

void updateCGPA(){

cout<<"Enter new CGPA: ";

cin>>cgpa; } void

updateHostel(){

cout<<"Enter new Hostel: ";

cin>>hostel; } void

displayDetails(){

cout<<name<<endl;

cout<<roll<<endl;

cout<<degree<<endl;

cout<<hostel<<endl;

cout<<cgpa<<endl;

}

};
```

```
int main(){ students  
  
s;  
  
s.addDetails();  
  
s.displayDetails();  
  
s.updateHostel();  
  
s.displayDetails();  
  
s.updateCGPA();  
  
s.displayDetails();  
  
return 0; }
```

```
soft-MIEngine-Pid-wnd3yuk5.mcd --dbgExe=C:\\msys64\\ucrt64\\bin\\gdb.exe --interpreter=mi  
tdin=Microsoft-MIEngine-In-zdoonzh4.fyp --stdout=Microsoft-MIEngine-Out-gt1oc0rt.tkn --stderr=Microsoft-MIEngine-Error-eknw5p5q.2x4 --pid=Micro  
soft-MIEngine-Pid-wnd3yuk5.mcd --dbgExe=C:\\msys64\\ucrt64\\bin\\gdb.exe --interpreter=mi  
Enter Name: mridul  
Enter Roll No: 1024150204  
tdin=Microsoft-MIEngine-In-zdoonzh4.fyp --stdout=Microsoft-MIEngine-Out-gt1oc0rt.tkn --stderr=Microsoft-MIEngine-Error-eknw5p5q.2x4 --pid=Micro  
soft-MIEngine-Pid-wnd3yuk5.mcd --dbgExe=C:\\msys64\\ucrt64\\bin\\gdb.exe --interpreter=mi  
Enter Name: mridul  
Enter Roll No: 1024150204  
Enter Name: mridul  
Enter Roll No: 1024150204  
Enter Degree: enc  
Enter Roll No: 1024150204  
Enter Degree: enc  
Enter Degree: enc  
Enter Hostel (Y/N): d  
Enter CGPA: 9.89  
  
Updating details...  
Enter new Name: mridul  
Enter new Roll No: 1024150204  
Enter new Degree: enc  
Enter new Hostel (Y/N): d  
Enter new CGPA: 9.58
```

Question:2 →

Public members can be accessed directly from outside the class using an object, while private members cannot be accessed directly and can only be used inside the class through member functions. This helps in data hiding and protects the internal data of the class from accidental modification.

```
#include <iostream>
```

```
#include <string> using
```

```
namespace std;
```

```
class Student {
```

```
private:
```

```
string name;
```

```
    int roll;
```

```
string degree;
```

```
string hostel;
```

```
float cgpa;
```

```
    void updateCGPA() {
```

```
cin >> cgpa;
```

```
    }
```

```
void updateHostel() {  
  
cin >> hostel;  
  
}  
  
public: void addDetails() {    cin >> name >> roll  
  
>> degree >> hostel >> cgpa;  
  
}  
  
void updateDetails() {  
  
updateCGPA();  
  
updateHostel();  
  
}  
  
void displayDetails() {  
  
cout << name << endl;  
  
cout << roll << endl;  
  
cout << degree << endl;  
  
cout << hostel << endl;  
  
cout << cgpa << endl;  
  
}  
  
};
```

```
int main() {  
  
    Student s;  
  
    s.addDetails();  
  
    s.updateDetails();  
  
    s.displayDetails();  
  
    return 0;  
  
}
```

```
$ /usr/bin/env c:\Users\devanshu\.vscode\extensions\ms-vscode.cpptools-1.29.3-win32-x64\debugAdapters\bin\WindowsDebugLauncher.exe --s  
tdin=Microsoft-MIEngine-In-2otrdcm.0rz --stdout=Microsoft-MIEngine-Out-evf5vm4d.b2f --stderr=Microsoft-MIEngine-Error-4ncozz5f.4vc --pid=Micro  
soft-MIEngine-Pid-slafwefe.ca5 --dbgExe=C:\msys64\ucrt64\bin\gdb.exe --interpreter=mi  
mridul  
1024151121  
enc  
d  
9  
9.1  
e  
mridul  
1024151121  
enc  
e  
9.1
```

Question:3 →

```
#include <iostream>  
  
using namespace std;  
  
class Demo { private:  
  
    void pvtFunction() { cout << "My  
  
    name is Vihaan!" << endl;
```

```
}
```

```
public:
```

```
void publicFunction() {
```

```
    pvtFunction();
```

```
}
```

```
};
```

```
int main() { Demo
```

```
    obj;
```

```
    obj.publicFunction()
```

```
; return 0;
```

```
}
```

```
$ /usr/bin/env c:\Users\devanshu\.vscode\extensions\ms-vscode.cpptools-1.29.3-win32-x64\debugAdapters\bin\WindowsDebugLauncher.exe --s
tdin=Microsoft-MIEngine-In-wdwwqbovg.ndt --stdout=Microsoft-MIEngine-Out-jdc33tuw.nja --stderr=Microsoft-MIEngine-Error-izriaaf.thu --pid=Micro
soft-MIEngine-Pid-w04cghlo.ues --dbgExe=C:\msys64\ucrt64\bin\gdb.exe --interpreter=mi
My name is Mridul!
```

Question:4 →

```
#include <iostream>
```

```
using namespace std;
```



```
class Rectangle { int  
  
width; int height;  
  
public: void getdata() {  
  
cin >> width >> height;  
  
}  
  
void calculatearea() { cout  
  
<< width * height;  
  
}  
  
};
```

```
int main() { Rectangle  
  
r;  
  
r.getdata();  
  
r.calculatearea();  
  
return 0; }
```

```
#include <iostream>  
  
using namespace std;  
  
class Rectangle { int  
  
width; int height;
```

public:

void getdata() { cin >>

width >> height;

}

void calculatearea() { cout

<< width * height;

}

};

int main() { Rectangle

r;

r.getdata();

r.calculatearea(); return

0;

}

```
$ /usr/bin/env c:\Users\devanshu\.vscode\extensions\ms-vscode.cpptools-1.29.3-win32-x64\debugAdapters\bin\windowsDebugLauncher.exe --s
tdin=Microsoft-MIEngine-In-dn3ganjh.ccv --stdout=Microsoft-MIEngine-Out-jg3wzija.xgx --stderr=Microsoft-MIEngine-Error-44sfqupo.unt --pid=Micro
soft-MIEngine-Pid-5qhfi5jp.dps --dbgExe=C:\msys64\ucrt64\bin\gdb.exe --interpreter=mi
soft-MIEngine-Pid-5qhfi5jp.dps --dbgExe=C:\msys64\ucrt64\bin\gdb.exe --interpreter=mi
10
10
100
```

Question:5 →

```
#include <iostream>
```

```
using namespace std;
```

```
class Complex {
```

```
float real; float
```

```
imag;
```

```
public: void setComplex(float r,
```

```
float i) { real = r; imag = i;
```

```
}
```

```
void displayComplex() { cout << real << " +
```

```
" << imag << "i" << endl;
```

```
}
```

```
Complex addComplex(Complex c) { Complex
```

```
temp;
```

```
temp.real = real + c.real;
```

```
temp.imag = imag + c.imag;
```

```
return temp;
```

```
}  
  
};  
  
int main() {  
  
Complex c1, c2, c3;  
  
c1.setComplex(2.5, 3.5);  
  
c2.setComplex(1.5, 2.5);  
  
c3 = c1.addComplex(c2);  
  
c3.displayComplex();  
  
return 0;  
  
}
```

```
$ /usr/bin/env c:\\Users\\devanshu\\.vscode\\extensions\\ms-vscode.cpptools-1.29.3-win32-x64\\debugAdapters\\bin\\WindowsDebugLauncher.exe --s  
tdin=Microsoft-MIEngine-In-h0i2wwut.ayu --stdout=Microsoft-MIEngine-Out-yu2z3bsc.0zw --stderr=Microsoft-MIEngine-Error-xea3a505.4aq --pid=Micro  
soft-MIEngine-Pid-seudv0x2.daq --dbgExe=C:\\msys64\\ucrt64\\bin\\gdb.exe --interpreter=mi  
4 + 6i
```

Q6 -

```
#include <iostream>  
  
using namespace std;
```

```
int x = 100;
```

```
class Demo { public:
```

```
    static int count;
```

```
    void show();
```

```
};
```

```
int Demo::count = 10;
```

```
void Demo::show() {
```

```
    int x = 50;    cout << "Local x: " << x << endl;    cout <<
```

```
"Global x: " << ::x << endl;    cout << "Static count: " <<
```

```
Demo::count << endl;    std::cout << "Using scope resolution
```

```
with cout" << std::endl;
```

```
}
```

```
int main() {
```

```
    Demo d;
```

```
    d.show();
```

```
    return 0;
```

```
}
```

```
$ /usr/bin/env c:\\Users\\devanshu\\.vscode\\extensions\\ms-vscode.cpptools-1.29.3-win32-x64\\debugAdapters\\bin\\WindowsDebugLauncher.exe --s
tdin=Microsoft-MIEngine-In-cou15lvd.050 --stdout=Microsoft-MIEngine-Out-qmi2uf35.1cf --stderr=Microsoft-MIEngine-Error-hcf4v0ja.mib --pid=Micro
soft-MIEngine-Pid-bfoptz4d.bb4 --dbgExe=C:\\msys64\\ucrt64\\bin\\gdb.exe --interpreter=mi
Local x: 50
Global x: 100
Static count: 10
Using scope resolution with cout
```

Q7 -

```
#include <iostream>
```

```
using namespace std;
```

```
namespace SectionA {
```

```
    int value = 10;    void display() {        cout <<
```

```
"SectionA value: " << value << endl;
```

```
    }
```

```
}
```

```
namespace SectionB {
```

```
    int value = 20;    void display() {        cout <<
```

```
"SectionB value: " << value << endl;
```

```
    }
```

```
}
```

```
int main() {
```

```
    SectionA::display();
```

```
SectionB::display();
```

```
return 0;
```

```
}
```

```
$ /usr/bin/env c:\\Users\\devanshu\\.vscode\\extensions\\ms-vscode.cpptools-1.29.3-win32-x64\\debugAdapters\\bin\\WindowsDebugLauncher.exe --s
tdin=Microsoft-MIEngine-In-xi3i4qn0.3ue --stdout=Microsoft-MIEngine-Out-34ibuerw.ukt --stderr=Microsoft-MIEngine-Error-otr2bnrv.f4x --pid=Micro
soft-MIEngine-Pid-nrdawnrr.gw2 --dbgExe=C:\\msys64\\ucrt64\\bin\\gdb.exe --interpreter=mi
SectionA value: 10
SectionB value: 20
```